

Pavan Bendre R

Bengaluru, Karnataka | pavanbr593@gmail.com | +91 82177 48699 | LinkedIn | GitHub | Portfolio

Professional Summary

Recent AI/ML graduate (VTU, 2026) with production-deployed experience in agentic pipelines, RAG systems, and government AI. Built and shipped a police scheduling system adopted by the Shivamogga Police Department, earned an IEEE Best Paper Award in healthcare AI, and published a technical book on IoT and Edge AI. Specializes in LLM orchestration, multi-agent systems, Hugging Face Transformers, and FastAPI-backed AI services across enterprise, healthcare, and public-sector domains.

Professional Experience

AI/ML Intern

Feb 2026 – Apr 2026

Saturam Inc., Bengaluru

- Built an end-to-end agentic partnerships-intelligence pipeline processing **400+ articles/run** across 12 enterprise partner feeds, eliminating **8+ hours/week** of manual research through fully automated LLM brief generation and email delivery.
- Engineered **multi-agent LLM orchestration** (LangChain + Groq Llama 3.3 70B, Gemini fallback) with rate-limit handling, retry logic, and **MiniLM + FAISS** semantic deduplication, cutting duplicate article noise by **~60%**.
- Delivered LLM-powered **recruitment intelligence agents** that reduced manual screening time by **40%** through automated resume scoring against structured JD criteria.

AI Engineer Intern

Nov 2024 – May 2025

OPQTECH Pvt Ltd, Remote

- Built a production-grade **multi-modal Retrieval-Augmented Generation (RAG)** platform combining document understanding, image embeddings, and local LLM reasoning to deliver grounded, citation-backed responses across PDFs and images.
- Implemented **CLIP + SentenceTransformers** embeddings with **FAISS** vector retrieval and **Phi-3** local inference, enabling explainable knowledge discovery with **zero cloud dependency** and sub-2s retrieval latency.

Key Projects

Police Bandobast Management System

React Native · FastAPI · Real-time Sync · Government Deployed

- AI-assisted duty-allocation system deployed for the Shivamogga Police Department, replacing conflict-prone manual scheduling with real-time, mobile-accessible workflows; **zero scheduling failures** post-deployment, earning an **official appreciation letter from the Superintendent of Police, Shivamogga**.

Illegal Waste Dumping Detection System

YOLOv8 · OpenCV · Computer Vision · Streamlit

- Developed a real-time computer vision system to detect illegal waste dumping from surveillance imagery, helping municipal authorities automate environmental monitoring and incident reporting.
- Trained and optimized a custom **YOLOv8** model on annotated waste-management datasets, integrating automated visual evidence logging and detection dashboards for actionable insights.

Non-Invasive Diabetes Prediction System

Regression Models · Gemini LLM · NIR Spectroscopy · Healthcare AI

- ML-based non-invasive diabetes prediction from physiological signals, with Gemini LLM generating automated health insights; **82% prediction accuracy**; extended into **IEEE Best Paper Award**-winning research published on IEEE Xplore.

Technical Skills

AI & GenAI: Agentic AI, Multi-Agent Orchestration, RAG, Conversational AI, Prompt Engineering, LLM Orchestration, LLM Evaluation

LLMs & Frameworks: LangChain, Llama 3.3 70B (Groq), Phi-3, Gemini, OpenAI API, Hugging Face Transformers, SentenceTransformers, CLIP

Retrieval & Vector: FAISS, Chroma, Pinecone, Embedding Pipelines, Semantic Search, Hybrid Search

ML / NLP / CV: TensorFlow, PyTorch, scikit-learn, Model Fine-Tuning (LoRA/QLoRA), MLOps, NER, Intent Classification, Computer Vision

Backend & DevOps: Python, FastAPI, Flask, REST APIs, Microservices, Docker, MongoDB, SQL, GCP, AWS, Azure, CI/CD (GitHub Actions), n8n

Frontend: React, React Native, Vite, Tailwind CSS, Gradio, TypeScript

Education

B.E. — Artificial Intelligence & Machine Learning

CGPA: 8.33

PES Institute of Technology and Management, Shivamogga (VTU, Belagavi)

Graduated June 2026

Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision

Best Outgoing Student Award

Publications & Awards

- IEEE Best Paper Award** — *Non-Invasive Estimation of Blood Glucose by Near-Infrared Spectroscopy and Machine Learning*, IEEE Xplore, 2025
- Published Author** (Amazon, 2024) — *Kickstart IoT Systems Engineering: Build Intelligent IoT Systems from Embedded Devices to Cloud and Edge AI*